

Mathilde Perez

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Education

PhD	Télécom Paris , Graph Learning, Fairness in AI - Fairness constraints on evolutionary graph learning models - Evaluation framework for biased vs. unbiased link prediction models	Palaiseau, France Sept 2023 – present
Master	Université Paul Sabatier , Applied Mathematics for Engineering, Industry and Innovation - AI, Deep Learning, Machine Learning, Big Data, image processing - Optimization, Statistics, Stochastic Simulations, PDEs, Signal Processing	Toulouse, France 2020 – 2022
Bachelor	Université Paul Sabatier , Applied Mathematics	Toulouse, France 2017 – 2020
Baccalauréat	Lycée des Arènes , Sciences Honors (Mention Bien), European English section	Toulouse, France 2016 – 2016

Publications

How Predicted Links Influence Network Evolution: Disentangling Choice and Algorithmic Feedback in Dynamic Graphs	2026
Mathilde Perez, Raphaël Romero, Jefrey Lijffijt, Charlotte Laclau arxiv.org/abs/2603.03945 (UAI 2026 · Spotlight presentation, ECAF 2026)	
TopoFair: Linking Topological Bias to Fairness in Link Prediction Benchmarks	2026
Lilian Marey, Mathilde Perez, Tiphaine Viard, Charlotte Laclau arxiv.org/abs/2602.11802 (Preprint)	
SimHawNet: a modified Hawkes process for temporal network simulation	2025
Mathilde Perez, Raphaël Romero, Bo Kang, Tijl De Bie, Jefrey Lijffijt, Charlotte Laclau 10.1007/s10618-025-01119-1 (Data Mining and Knowledge Discovery)	

Experience

Télécom Paris , Research Engineer Simulation of temporal graphs with Hawkes processes	Palaiseau, France Apr 2023 – Sept 2023 6 months
ISIR — Sorbonne / CNRS , Human-Computer Interaction Intern Modeling social attention through gaze in a conversational agent (Greta platform)	Paris, France Apr 2022 – Oct 2022 7 months
CerCo — CNRS , NLP Intern Similarities between CLIP and the brain	Purpan, France May 2021 – July 2021 3 months

Oral Presentations

- SimHawNet: a modified Hawkes process for temporal network simulation** — ECML-PKDD 2025, Porto, Portugal · [Slides](#) & CAp 2024, Paris, France
- How social media algorithms reinforce inequality** — [EUNIC Science Day](#), Madrid, Spain · [Slides](#)

Skills

Programming

Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, C/C++/C#, R, LaTeX, Unity, Java

Machine Learning & AI

Graph Learning, Deep Learning, Fairness in AI, Temporal Graphs, Hawkes processes

Soft Skills

Teamwork, Public speaking, Autonomy

Languages

French

Native

English

Advanced